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MATHS MASTER _____

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CANDIDATE NUMBER

2023 Trial HSC Examination

Form VI Mathematics Extension 2

Tuesday 8th August 2023

8:40am

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Total Marks: 100

Section I (10 marks) Questions 1 – 10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11 – 16

- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on each booklet and on the multiple choice sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.
- Place everything inside this question booklet.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 6 booklets per boy
- Candidature: 81 pupils

Writer: RR

Section I

Questions in this section are multiple-choice.

Record the single best answer for each question on the provided answer sheet.

1. Which of the following is the correct expression for $\int \frac{1}{\sqrt{1-9x^2}} dx$?

(A) $\frac{1}{3} \sin^{-1} \frac{x}{3} + C$

(B) $3 \sin^{-1} \frac{x}{3} + C$

(C) $\frac{1}{3} \sin^{-1} 3x + C$

(D) $3 \sin^{-1} 3x + C$

2. If $\underline{a}$ and $\underline{b}$ satisfy $(\underline{a} - \underline{b}) \cdot (\underline{a} + \underline{b}) = 0$, which of the following must be true?

(A) $\underline{a} = \pm \underline{b}$

(B) $|\underline{a}| = |\underline{b}|$

(C) $\underline{a}$ is parallel to $\underline{b}$

(D) $\underline{a}$ is perpendicular to $\underline{b}$

3. Consider the statement:

$$\forall a \in \mathbf{Z}, \exists b \in \mathbf{Z} \text{ such that } \frac{a}{b} \in \mathbf{Z}$$

What does this statement mean?

(A) Every integer has a factor.

(B) There exists an integer which divides every integer.

(C) Every integer is also a rational number.

(D) There exists a rational number corresponding to each pair of integers.

4. Which of the following is the correct expression for $\int x \cos x dx$?

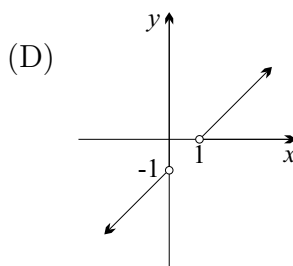
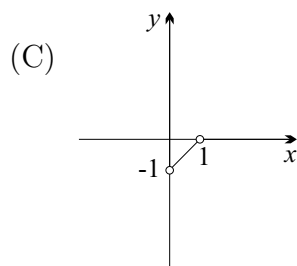
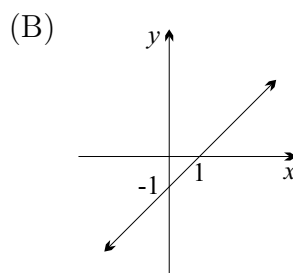
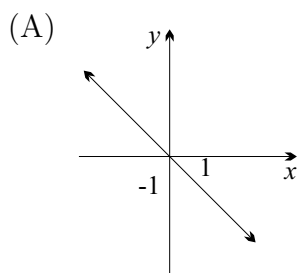
(A) $x \sin x - \sin x + C$

(B) $x \sin x + \cos x + C$

(C) $x \cos x - \cos x + C$

(D) $x \cos x + \sin x + C$

5. Which of the following curves satisfies the equation $\text{Arg}(z - 1) = \text{Arg}(z + i)$?



6. Consider the statement:

‘If n is even, then $n^3 + n$ is even.’

Which of the following are true?

- (A) The contrapositive of the original statement but NOT the converse
- (B) The converse of the original statement but NOT the contrapositive
- (C) Both the contrapositive and the converse of the original statement
- (D) Neither the contrapositive nor the converse of the original statement

7. What is the angle made between the line $\mathcal{L} = \lambda \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, where $\lambda \in \mathbf{R}$, and the yz -plane?

You may assume $a, b, c > 0$.

- (A) $\sin^{-1} \left(\frac{a}{\sqrt{a^2 + b^2 + c^2}} \right)$
- (B) $\sin^{-1} \left(\frac{b}{\sqrt{a^2 + b^2 + c^2}} \right)$
- (C) $\tan^{-1} \left(\frac{b}{\sqrt{a^2 + c^2}} \right)$
- (D) $\tan^{-1} \left(\frac{c}{\sqrt{a^2 + b^2}} \right)$

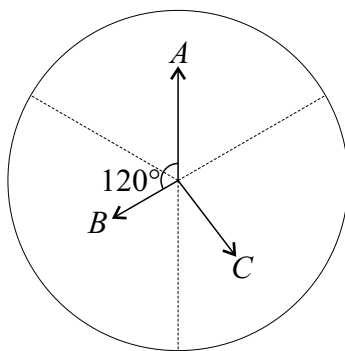
8. Which of the following gives the correct expression for $\int \frac{\sin x}{\sin x + \cos x} dx$?

- (A) $\frac{1}{2}(x + \ln |\sin x + \cos x|) + C$
- (B) $\frac{1}{2}(x - \ln |\sin x + \cos x|) + C$
- (C) $-\frac{1}{2}(x + \ln |\sin x + \cos x|) + C$
- (D) $-\frac{1}{2}(x - \ln |\sin x + \cos x|) + C$

9. If z is a complex number such that $|z + i| = 5$, what is the maximum value of $|z^2 - 1|$?

- (A) 17
- (B) 25
- (C) 35
- (D) 37

10. Alpha, Bravo, and Charlie are competing in a three-way tug-of-war. They each hold a rope, and the three ropes are joined together at the centre of a circle. A competitor wins if they can pull the intersection point into their sector (see below):



Alpha and Bravo pull with a force of a and b newtons respectively at an angle of 120° to one another. Charlie pulls with a force of c newtons and pulls at an angle directly opposing the sum of the two other forces.

Given this scenario, which of the following statements is true?

- (A) If $c > \frac{a+b}{2}$, then Charlie wins.
- (B) If $c > ab$, then Charlie wins.
- (C) If Charlie wins, then $c^2 > ab$.
- (D) If Charlie wins, then $c > a + b$.

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet.

Marks

(a) Let $z = \sqrt{3} + i$ and $w = 1 - i$.

(i) Express $\frac{z}{w}$ in the form $a + bi$.

1

(ii) Express z and w in modulus-argument form.

1

(iii) Hence determine the exact value of $\tan \frac{5\pi}{12}$.

2

(b) Evaluate $\int_0^1 \frac{4x+1}{x+1} dx$.

2

(c) Consider the statement:

2

‘If $n^2 - 3n + 2$ is odd, then n is odd.’

Prove that the statement is true using contraposition.

(d) (i) Use the result $e^{i\theta} = \cos \theta + i \sin \theta$ to show $e^{ni\theta} + e^{-ni\theta} = 2 \cos n\theta$.

1

(ii) By expanding $(e^{i\theta} + e^{-i\theta})^4$, prove that $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$.

2

(iii) Hence find $\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$.

1

(e) A particle experiences acceleration according to the equation $\ddot{x} = 2v$, where v is the particle’s velocity in metres per second. It begins at the origin with a velocity of 1 m/s.

3

How far does it travel in the first two seconds? Write your answer correct to the nearest metre.

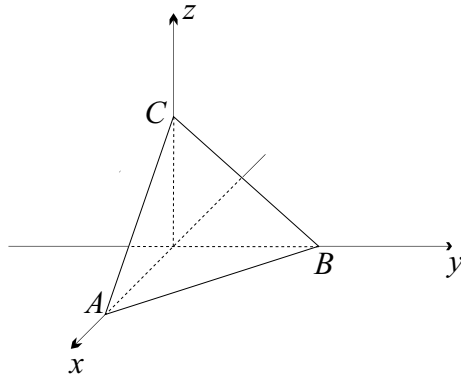
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QUESTION TWELVE (15 marks)

Start a new answer booklet.

Marks

- (a) Let ABC be a triangle with vertices $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, k)$ for some constant $k > 0$. This is shown below:



- (i) Express $\overrightarrow{AC}$ and $\overrightarrow{BC}$ as column vectors. 1
- (ii) Find the value of k that will make $\angle ACB = 45^\circ$. Write your answer correct to 2 decimal places. 2
- (b) (i) Find the constants A , B and C such that $\frac{x^2 + 8x + 16}{x^3 + 4x} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$. 2
- (ii) Hence find $\int \frac{x^2 + 8x + 16}{x^3 + 4x} dx$. 2
- (c) The acceleration of an object moving in a straight line is given by $\ddot{x} = -\frac{1}{1+x}$, where x metres is the displacement of the object from the origin. The object is initially at the origin with a velocity of 2 m/s.
- (i) Find v^2 as a function of x . 2
- (ii) Hence find where the object will first come to a rest, and justify whether it must turn back around or not. 2
- (d) (i) Prove using contradiction that $\sqrt{6}$ is irrational. 2
- (ii) Hence prove that it is not possible for both $\sqrt{2n}$ and $\sqrt{3n}$ to be rational for any positive integer n . 2

The paper continues on the next page

QUESTION THIRTEEN (15 marks)

Start a new answer booklet.

Marks(a) Consider the points $A(5, 0, 2)$, $B(2, 3, -4)$ and $C(8, -3, 5)$.(i) Find an equation for the line AB .

1

(ii) Show that the point C is not on the line AB .

1

(iii) Find the point on the line AB which is closest to point C .

2

(b) A particle moves in a straight line and its motion satisfies the equation

$$v^2 = 32 + 8x - 4x^2,$$

where x is the particle's displacement from the origin in metres, and v is its velocity in metres per second.

(i) Show that the motion is simple harmonic.

1

(ii) Find the maximum acceleration $|\ddot{x}|$ achieved by the particle.

2

(c) Use the substitution $u = \sqrt{e^x - 1}$ to evaluate $\int_0^{\ln 2} \sqrt{e^x - 1} \, dx$.

3

(d) (i) Use deMoivre's Theorem to show that

2

$$\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.$$

(ii) By considering the solutions of $\sin 5\theta = 0$, find the exact value of $\sin^2 \frac{\pi}{5}$.

3

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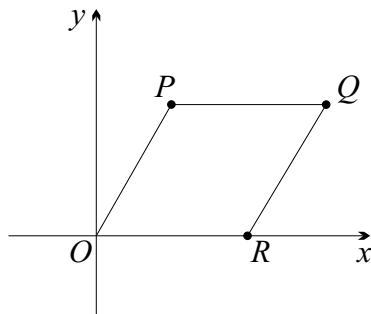
QUESTION FOURTEEN (15 marks)

Start a new answer booklet.

Marks

- (a) Let $z = \cos \theta + i \sin \theta$ be some complex number with $0 < \theta < \frac{\pi}{2}$ and let the points P , Q , and R represent the complex numbers z , $z + 1$, and 1 respectively, as shown below:

4



By considering the quadrilateral $OPQR$, or otherwise, show that

$$\frac{2z}{z+1} = 1 + i \tan \frac{\theta}{2}.$$

- (b) Let x and y be positive real numbers.

(i) Prove that $x + \frac{1}{x} \geq 2$, for all $x > 0$.

1

(ii) Hence show that

2

$$\frac{1+x^2}{y} + \frac{1+y^2}{x} \geq 4.$$

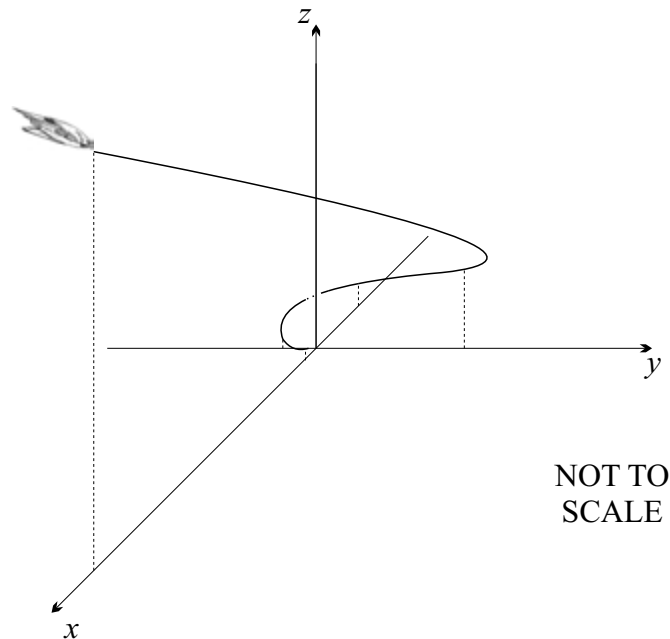
- (c) By using the substitution $t = \tan \frac{x}{2}$, find $\int \frac{1}{1 + \sin x - \cos x} dx$.

3

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QUESTION FOURTEEN (Continued)

- (d) A trained falcon is soaring above a crowd before gliding down to land on its trainer. In order to maintain constant eye contact without having to move its eyes or head, it glides along a path known as a logarithmic spiral, as shown below:



The position of the falcon, t seconds after beginning its descent, is given by

$$\underline{r} = \begin{pmatrix} 20e^{-t} \cos t \\ 20e^{-t} \sin t \\ 10e^{-t} \end{pmatrix},$$

where the units of the components are in metres.

- (i) Find an expression for the velocity $\underline{v}$ of the falcon at any time t , and hence calculate its speed as it begins its descent. 2

- (ii) Show that the angle between $\underline{r}$ and $\underline{v}$ is constant. 3

The paper continues on the next page

QUESTION FIFTEEN (16 marks)

Start a new answer booklet.

Marks

- (a) (i) Find the three cube roots of $2 + 2i$ and plot these roots on the Argand diagram. **3**

- (ii) Given that the three roots sum to zero, show that **2**

$$\cos\left(\frac{\pi}{12}\right) - \sin\left(\frac{\pi}{12}\right) = \frac{1}{\sqrt{2}}.$$

- (b) Consider the polynomial $P(z) = z^4 + kz^2 + 1$, for some $k \in \mathbf{R}$.

- (i) Show that if $z = \omega$ is a zero of $P(z)$, then so are $-\omega$, $\overline{\omega}$, and ω^{-1} . **2**

- (ii) If $P(\omega) = 0$ and $|\omega| \neq 1$, justify that either $\operatorname{Re}(\omega) = 0$ or $\operatorname{Im}(\omega) = 0$. **3**

- (c) The HMS Dreadnought was a British battleship in the early 20th century and is the only battleship to have ever sunk a submarine. As part of its artillery, it could launch torpedoes of mass 500 kg at an initial speed of 10 m/s into the water. While in the water, the torpedo would self-propel with a constant driving force of 2000 N and approach a terminal velocity of 20 m/s.

For this question, you can assume that the torpedo travels horizontally through the water at a constant depth and that the torpedo experiences resistance from the water proportional to the square of its velocity.

- (i) Show that the displacement of the torpedo from the ship satisfies the equation **2**

$$\ddot{x} = 4 - \frac{v^2}{100}.$$

- (ii) How long would it take the torpedo to hit a target 1000 m away? Write your answer correct to the nearest second. **4**

The paper continues on the next page

QUESTION SIXTEEN (14 marks) Start a new answer booklet.**Marks**

- (a) Let
- n
- be a positive odd integer and define recursively the double factorial:

3

$$n!! = \begin{cases} n \times (n-2)!!, & \text{for } n \geq 3 \\ 1, & \text{for } n = 1 \end{cases}$$

Note that $n!!$ is not the same as $(n!)!$.

Prove using induction that

$$n!! = \frac{n!}{2^{\frac{n-1}{2}} \times \left(\frac{n-1}{2}\right)!}$$

for all positive odd integers.

- (b) Consider the definite integral

$$I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx,$$

where m and n are non-negative integers.

- (i) Using the substitution
- $x = \frac{\pi}{2} - u$
- , show that
- $I_{m,n} = I_{n,m}$
- .

1

- (ii) Prove that
- $I_{m,n} = \frac{n-1}{m+n} I_{m,n-2}$
- , for
- $n \geq 2$
- .

3

- (iii) Using the previous results from both part (a) and part (b), show that

4

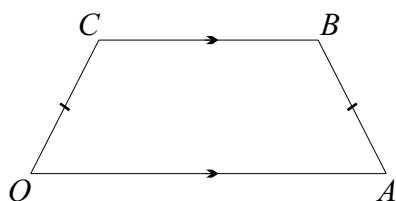
$$\int_0^{\frac{\pi}{2}} \sin^{2k} x \cos^{2l} x \, dx = \frac{(2k)!(2l)!}{k!l!(k+l)!} \times \frac{\pi}{2^{2k+2l+1}}$$

for all non-negative integers k and l .

- (c) An isosceles trapezium
- $OABC$
- is a quadrilateral with sides
- OA
- and
- BC
- parallel but NOT equal in length, and sides
- OC
- and
- AB
- equal in length but not parallel.

3

This is shown below:



Let $\underline{a} = \overrightarrow{OA}$ and $\underline{c} = \overrightarrow{OC}$.

Show, using vector methods, that the diagonals OB and AC are equal in length.

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SYDNEY GRAMMAR SCHOOL



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Form VI Mathematics Extension 2

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- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

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Extension 2 2023 Trial - Solutions

Q1. C Note $\frac{d}{dx} \sin^{-1}(3x) = \frac{3}{\sqrt{1-(3x)^2}}$

Q2. B $(a-b) \cdot (a+b) = |a|^2 - |b|^2 = 0$

Q3. A "For all a, there is a b which divides it."

Q4. B $\frac{d}{dx} (x \sin x + \cos x) = x \cos x - \sin x + \sin x$

Q5. D Some direction from 1 and -i.

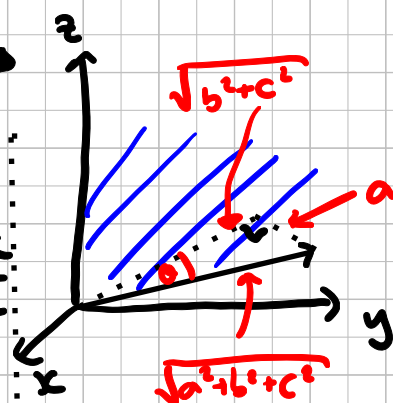
Q6. A The original statement is true but not the converse

Q7. A See diagram to right

Q8. B Best done by testing...

$$\frac{d}{dx} (\ln |\sin x + \cos x|) = \frac{\cos x - \sin x}{\sin x + \cos x}$$

$$= 1 - 2 \frac{\sin x}{\cos x}$$

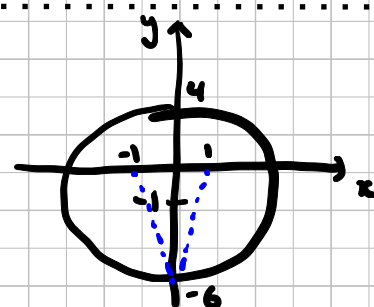


Q9. D $|z+i|=5$ so z sits on circle:

$$|z^2-1| = |z+1||z-1|$$

which is maximised by $z=6i$

$$= \sqrt{37} \times \sqrt{37}$$



Q10. C The others are easily seen to be false by example:

~~A~~ If $a=100, b=0, c=51$: Charlie should lose

~~B~~ As above

~~D~~ If $a=100, b=100, c=101$: Charlie should win

To see why C is true: Charlie wins precisely when $c^2 > a^2 + b^2 - 2ab \cos 60^\circ$
 $= a^2 + b^2 - ab$
 $= (a-b)^2 + ab$

Hence if Charlie wins, $c^2 > ab$

Q11 a) i) $z = \sqrt{3} + i$, $w = 1 - i$

$$\begin{aligned}\frac{z}{w} &= \frac{\sqrt{3} + i}{1 - i} \times \frac{1 + i}{1 + i} \\ &= \frac{(\sqrt{3} - 1) + (\sqrt{3} + 1)i}{2} \\ &= \frac{\sqrt{3} - 1}{2} + \frac{\sqrt{3} + 1}{2}i \quad \checkmark\end{aligned}$$

ii) $z = 2 \operatorname{cis} \frac{\pi}{6}$, $w = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)$ $\checkmark$

iii) $\frac{z}{w} = \frac{2}{\sqrt{2}} \operatorname{cis} \left(\frac{\pi}{6} + \frac{\pi}{4} \right)$
 $= \frac{2}{\sqrt{2}} \operatorname{cis} \left(\frac{5\pi}{12} \right) \quad \checkmark$

Equating real parts: $\frac{2}{\sqrt{2}} \cos \frac{5\pi}{12} = \frac{\sqrt{3} - 1}{2}$ (2)

Equating imaginary parts: $\frac{2}{\sqrt{2}} \sin \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{2}$ (1)

(1) $\div$ (2): $\tan \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ $\checkmark$
 $= 2 + \sqrt{3}$

11 b) $\int_0^1 \frac{4x+1}{x+1} dx = \int_0^1 \frac{4x+4}{x+1} - \frac{3}{x+1} dx$
 $= \left[4x - 3 \ln|x+1| \right]_0^1 \quad \checkmark$
 $= 4 - 3 \ln 2 \quad \checkmark$

11c) The contrapositive statement is
'If n is even, then $n^2 - 3n + 2$ is even'
(✓ For clear reference/understanding of contrapositive)

So suppose $n = 2k$ for some $k \in \mathbb{Z}$.

$$\begin{aligned}\text{Then } n^2 - 3n + 2 &= (2k)^2 - 3(2k) + 2 \\ &= 4k^2 - 6k + 2 \\ &= 2(2k^2 - 3k + 1) \quad \checkmark\end{aligned}$$

which is even.

Hence, by contraposition, the original statement is true.

$$\begin{aligned}\text{11d)i) } e^{in\theta} + e^{-in\theta} &= [\cos(n\theta) + i \sin(n\theta)] \\ &\quad + [\cos(-n\theta) + i \sin(-n\theta)] \quad \checkmark \\ &= \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta) \\ &\quad \left\{ \begin{array}{l} \text{as } \cos\theta \text{ even} \\ \text{ \& } \sin\theta \text{ odd} \end{array} \right\} \\ &= 2\cos n\theta\end{aligned}$$

$$\begin{aligned}\text{ii) } (e^{i\theta} + e^{-i\theta})^4 &= e^{i4\theta} + 4e^{i2\theta} + 6 + 4e^{-i2\theta} + e^{-i4\theta} \quad \checkmark \\ &= (e^{i4\theta} + e^{-i4\theta}) + 4(e^{i2\theta} + e^{-i2\theta}) + 6\end{aligned}$$

$$\text{so } (2\cos\theta)^4 = 2\cos 4\theta + 4(2\cos 2\theta) + 6 \quad \checkmark$$

$$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$$

$$\Rightarrow \cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$$

$$\begin{aligned}
 11 d) \text{ iii) } \int_0^{\frac{\pi}{2}} \cos^4 \theta &= \int_0^{\frac{\pi}{2}} \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \\
 &= \left[\frac{1}{32} \sin 4\theta + \frac{1}{4} \sin 2\theta + \frac{3}{8} x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{3\pi}{16} \checkmark
 \end{aligned}$$

11 e)

$$\ddot{x} = 2v$$

$$\frac{dv}{dt} = 2v$$

$$\frac{dt}{dv} = \frac{1}{2v}$$

integrating, $t = \frac{1}{2} \ln|v| + C$
 when $t=0, v=1$ so $C=0$

$$t = \frac{1}{2} \ln(v) \checkmark$$

$$v = e^{2t}$$

$$\frac{dx}{dt} = e^{2t}$$

integrating, $x = \frac{1}{2} e^{2t} + C \checkmark$

when $t=0, x=0$ so $C = -\frac{1}{2}$

$$x = \frac{1}{2} (e^{2t} - 1)$$

letting $t=2$,

$$x = 2.7 \text{ m} \checkmark \text{ (nearest metre)}$$

alt:

$$v \frac{dv}{dx} = 2v$$

$$\frac{dv}{dx} = 2$$

$$v = 2x + C$$

when $x=0, v=1$

so $C=1$

$$v = 2x + 1$$

$$\frac{dx}{dt} = 2x + 1$$

$$t = \frac{1}{2} \ln(2x+1) + C$$

when $x=0, t=0$

so $C=0$

$$\Rightarrow x = \frac{1}{2} (e^{2t} - 1)$$

12a) i) $\vec{AC} = \begin{pmatrix} -1 \\ 0 \\ k \end{pmatrix}$ and $\vec{BC} = \begin{pmatrix} 0 \\ -1 \\ k \end{pmatrix}$ ✓

ii) If $\angle ACB = 45^\circ$, then

$$\frac{\vec{AC} \cdot \vec{BC}}{|\vec{AC}| |\vec{BC}|} = \cos 45^\circ$$

so $\frac{k^2}{\sqrt{k^2+1} \cdot \sqrt{k^2+1}} = \frac{1}{\sqrt{2}}$ ✓

$\Rightarrow \sqrt{2} k^2 = k^2 + 1$

$(\sqrt{2} - 1) k^2 = 1$

$k^2 = \frac{1}{\sqrt{2} - 1}$

$k \approx 1.55$ ✓

b) i) let $\frac{x^2+8x+16}{x^3+4x} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$

then $x^2+8x+16 = A(x^2+4) + (Bx+C)x$

Equating constants: $4A = 16 \Rightarrow A = 4$ ✓

Equating coefficients of x : $C = 8$

Equating coefficients of x^2 : $A+B = 1 \Rightarrow B = -3$ } ✓

ii) $\int \frac{x^2+8x+16}{x^3+4x} dx = \int \frac{4}{x} - \frac{3x}{x^2+4} + \frac{8}{x^2+4} dx$

$= 4 \ln|x| - \frac{3}{2} \ln|x^2+4|$ ✓

$+ 4 \tan^{-1} \frac{x}{2} + C$ ✓

12c) i)

$$\ddot{x} = -\frac{1}{1+x}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -\frac{1}{1+x}$$

$$\frac{1}{2}v^2 = -\ln|1+x| + \frac{C}{2} \quad \checkmark$$

$$v^2 = C - 2\ln|1+x|$$

when $x=0$, $v=2$ so $C=4$

$$v^2 = 4 - 2\ln|1+x| \quad \checkmark$$

ii) Letting $v^2=0$ gives

$$2\ln|1+x| = 4$$

$$1+x = e^2 \quad (\text{note } v>0 \text{ initially so } x>0)$$

$$x = e^2 - 1 \quad \checkmark$$

when $x = e^2 - 1$

$$\ddot{x} = -\frac{1}{e^2}$$

$$< 0$$

so the object will turn around as v becomes negative. $\checkmark$

12d) i) Suppose, by way of contradiction, that $\sqrt{6}$ is rational. $\checkmark$

I.e. $\sqrt{6} = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with no common factors.

$$\text{Then } 6 = \frac{a^2}{b^2}$$

$$\text{so } 6b^2 = a^2$$

Since LHS is even, RHS is even and hence a is even. $\checkmark$

$$\text{Let } a = 2k$$

$$\begin{aligned} \text{so } 6b^2 &= (2k)^2 \\ &= 4k^2 \end{aligned}$$

$$\Rightarrow 3b^2 = 2k^2$$

12 di) (cont.) $3b^2 = 2k^2$

Now RHS is even, so LHS is even $\Rightarrow b$ is even.

But then a & b are both even, so they share a common factor, which is a contradiction. ✓

Thus, by contradiction, $\sqrt{6}$ is irrational.

ii) Now suppose, by way of contradiction, that both $\sqrt{2n}$ and $\sqrt{3n}$ are rational.

I.e. $\sqrt{2n} = \frac{p}{q}$ and $\sqrt{3n} = \frac{r}{s}$ ✓ for $p, q, r, s \in \mathbb{Z}$

but then $\sqrt{2n} \times \sqrt{3n} = \frac{pr}{qs}$

$$\sqrt{6} n^2 = \frac{pr}{qs}$$

$$\sqrt{6} = \frac{pr}{n^2 qs}$$

which implies $\sqrt{6}$ is rational. ✓

This is a contradiction of part i), and hence $\sqrt{2n}$ and $\sqrt{3n}$ cannot both be rational.

13a)

$$A(5, 0, 2), B(2, 3, -4), C(8, -3, 5)$$

$$\begin{aligned} \text{i) } \underline{r} &= \vec{OA} + \lambda \vec{AB} \\ &= \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix} \checkmark \end{aligned}$$

$$\text{ii) Suppose } \begin{pmatrix} 8 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix}$$

$$\text{then } 8 = 5 - 3\lambda \Rightarrow \lambda = -1$$

but equating z coefficients gives

$$5 = 2 - 6\lambda$$

$$\Rightarrow 6\lambda = -3$$

$$\Rightarrow \lambda = -\frac{1}{2} \checkmark$$

since these do not agree, C is not on AB.

$$\text{iii) } \vec{AC} = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$$

$$\begin{aligned} \text{proj}_{\vec{AB}} \vec{AC} &= \frac{\begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix}}{\left| \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix} \right|^2} \cdot \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix} \\ &= \frac{-36}{54} \begin{pmatrix} -3 \\ 3 \\ -6 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix} \checkmark \end{aligned}$$

$$\text{so closest point is } \vec{OA} + \begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -2 \\ 6 \end{pmatrix} \checkmark$$

13b) i) $v^2 = 32 + 8x - 4x^2$

$$\frac{1}{2} v^2 = 16 + 4x - 2x^2$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4 - 4x$$

$$\ddot{x} = -4(x-1) \checkmark \text{ (must be factorised)}$$

Hence the motion is simple harmonic
(with centre of motion $x=1$)

ii) Let $v=0$, then

$$32 + 8x - 4x^2 = 0$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow (x-4)(x+2) = 0$$

so the particle is at rest when $x=-2$ or $x=4$ $\checkmark$

At these points $|\ddot{x}| = 4|x-1|$
 $= 12$

so max acceleration is 12 m/s^2 . $\checkmark$

13 c). $\int_0^{\ln 2} \sqrt{e^x - 1} \, dx$

$$= \int_0^1 u \cdot \frac{2u}{u^2+1} \, du$$

$$= \int_0^1 2 - \frac{2}{u^2+1} \, du \checkmark$$

$$= \left[2u - 2 \tan^{-1} u \right]_0^1$$

$$= 2 - \frac{\pi}{2} \checkmark$$

let $u = \sqrt{e^x - 1}$

so $du = \frac{e^x}{2\sqrt{e^x - 1}} \, dx \checkmark$

$$= \frac{u^2+1}{2u} \, dx$$

so $\frac{2u}{u^2+1} \, du = dx$

when $x=0$, $u=0$
 $x=\ln 2$, $u=1$

$$13d) i) (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta \quad \{\text{De Moivre's Theorem}\}$$

$$\text{and } (\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \quad \checkmark$$

Equating imaginary parts:

$$\begin{aligned} \sin 5\theta &= 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \\ &= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \quad \checkmark \\ &= 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta \quad \text{as req'd.} \end{aligned}$$

$$ii) \text{ Note that } \sin \left(5 \times \frac{\pi}{5} \right) = \sin \pi = 0$$

$$\text{so } 5 \sin \frac{\pi}{5} - 20 \sin^3 \frac{\pi}{5} + 16 \sin^5 \frac{\pi}{5} = 0$$

clearly $\sin \frac{\pi}{5} \neq 0$, so divide through to get

$$16 \sin^4 \frac{\pi}{5} - 20 \sin^2 \frac{\pi}{5} + 5 = 0 \quad \checkmark$$

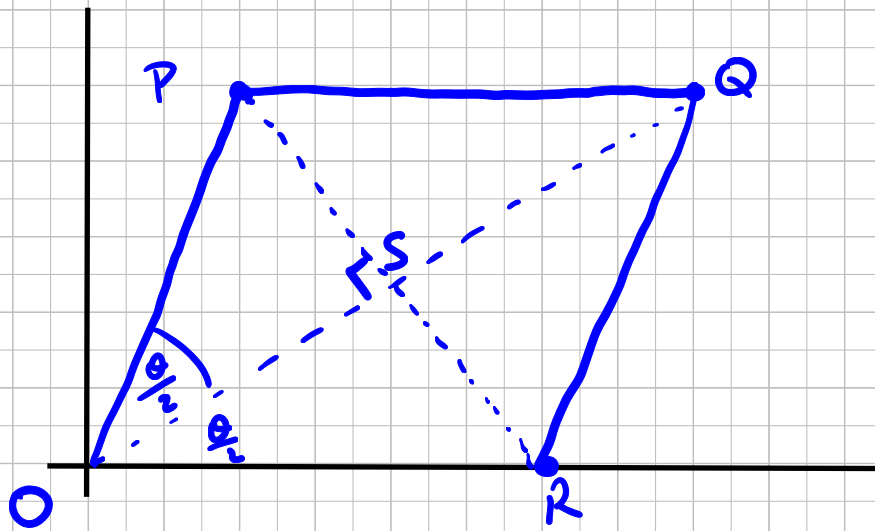
$$16 \left(\sin^2 \frac{\pi}{5} \right)^2 - 20 \left(\sin^2 \frac{\pi}{5} \right) + 5 = 0$$

$$\begin{aligned} \text{so } \sin^2 \frac{\pi}{5} &= \frac{20 \pm \sqrt{400 - 320}}{32} \quad \checkmark \\ &= \frac{5 \pm \sqrt{5}}{8} \end{aligned}$$

$$\text{and note } \sin^2 \frac{\pi}{5} < \sin^2 \frac{\pi}{4} = \frac{1}{2}$$

$$\text{so } \sin^2 \frac{\pi}{5} = \frac{5 - \sqrt{5}}{8} \quad \checkmark$$

14a)



Note that $|z|=1$ so $OPQR$ is a rhombus as it is a parallelogram with adjacent sides equal.

Hence $\arg(z+1) = \frac{\theta}{2}$ ✓ (diagonal bisects vertex)

$$\begin{aligned} \text{so } \arg\left(\frac{2z}{z+1}\right) &= \arg(2z) - \arg(z+1) \\ &= \theta - \frac{\theta}{2} \\ &= \frac{\theta}{2}. \quad \checkmark \end{aligned}$$

and letting OQ & PR intersect at S
 we know $OQ \perp PR$
 and OQ bisects PR } (diagonals of a rhombus)

$$\text{so } |OS| = \cos \frac{\theta}{2}$$

$$\Rightarrow |z+1| = 2 \cos \frac{\theta}{2} \quad \checkmark$$

$$\begin{aligned} \text{hence } \left| \frac{2z}{z+1} \right| &= \frac{|2z|}{|z+1|} \\ &= \frac{1}{\cos \frac{\theta}{2}} \end{aligned}$$

$$\begin{aligned} \text{so } \frac{2z}{z+1} &= \frac{1}{\cos \frac{\theta}{2}} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right) \quad \checkmark \\ &= 1 + i \tan \frac{\theta}{2} \quad \text{as required.} \end{aligned}$$

14b) i) RTP: $x + \frac{1}{x} \geq 2$ for all $x > 0$

$$\begin{aligned} \text{LHS} - \text{RHS} &= x - 2 + \frac{1}{x} \\ &= \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \checkmark \\ &\geq 0 \end{aligned}$$

Hence $\text{LHS} - \text{RHS} \geq 0$
 $\Rightarrow \text{LHS} \geq \text{RHS}$

ii) Since $x + \frac{1}{x} \geq 2$, it follows

$$x^2 + 1 \geq 2x \quad \text{as } x > 0.$$

$$\begin{aligned} \text{Hence } \frac{1+x^2}{y} + \frac{1+y^2}{x} &\geq \frac{2x}{y} + \frac{2y}{x} \checkmark \\ &= 2\left(\frac{x}{y} + \frac{y}{x}\right) \\ &\geq 2 \times 2 \checkmark \quad (\text{from part i}) \\ &= 4. \end{aligned}$$

Q14c). $\int \frac{1}{1 + \sin x - \cos x} dx$

Let $t = \tan \frac{x}{2}$

$$= \int \frac{1}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \checkmark$$

$$\begin{aligned} 2 \tan^2 t &= x \\ \frac{2 dt}{1+t^2} &= dx \end{aligned}$$

$$= \int \frac{2 dt}{2t + 2t^2}$$

$$= \int \frac{1}{t + t^2} dt$$

$$= \int \left(\frac{1}{t} - \frac{1}{1+t} \right) dt \checkmark$$

$$= \ln |t| - \ln |1+t| + C$$

$$= \ln \left(\tan \frac{x}{2} \right) - \ln \left(1 + \tan \frac{x}{2} \right) + C \checkmark$$

14 d)

$$\vec{r} = \begin{pmatrix} 20e^{-t} \cos t \\ 20e^{-t} \sin t \\ 10e^{-t} \end{pmatrix}$$

$$\vec{v} = \begin{pmatrix} -20e^{-t} \cos t - 20e^{-t} \sin t \\ -20e^{-t} \sin t + 20e^{-t} \cos t \\ -10e^{-t} \end{pmatrix} \quad \checkmark$$

$$\text{when } t=0, \quad \vec{v} = \begin{pmatrix} -20 \\ 20 \\ -10 \end{pmatrix}$$

$$|\vec{v}| = 30 \text{ m/s} \quad \checkmark$$

$$\begin{aligned} \text{ii) } \vec{r} \cdot \vec{v} &= (-400e^{-2t} \cos^2 t - 400e^{-2t} \sin t \cos t) \\ &\quad + (-400e^{-2t} \sin^2 t + 400e^{-2t} \sin t \cos t) \\ &\quad + (-100e^{-2t}) \\ &= -500e^{-2t} \quad \checkmark \end{aligned}$$

$$\begin{aligned} |\vec{r}|^2 &= 400e^{-2t} \cos^2 t + 400e^{-2t} \sin^2 t + 100e^{-2t} \\ &= 500e^{-2t} \end{aligned}$$

$$|\vec{r}| = 10\sqrt{5} e^{-t}$$

$$\begin{aligned} |\vec{v}|^2 &= (400e^{-2t} \cos^2 t + 800e^{-2t} \cos t \sin t + 400e^{-2t} \sin^2 t) \\ &\quad + (400e^{-2t} \sin^2 t - 800e^{-2t} \cos t \sin t + 400e^{-2t} \cos^2 t) \\ &\quad + 100e^{-2t} \end{aligned}$$

14d) ii) (cont.)

$$|\underline{x}|^2 = 900 e^{-2t}$$

$$|\underline{v}| = 30 e^{-t} \quad \checkmark$$

Let θ be the angle between $\underline{r}$ & $\underline{x}$

$$\text{then } \cos \theta = \frac{\underline{r} \cdot \underline{v}}{|\underline{r}| |\underline{v}|}$$

$$= \frac{-500 e^{-2t}}{(10\sqrt{5} e^{-t})(30 e^{-t})}$$

$$\cos \theta = \frac{-\sqrt{5}}{3} \quad \checkmark$$

$$\hat{=} 138^\circ$$

(The falcon looks at a constant angle of 42°).

15a) i)

Let $z = r \operatorname{cis} \theta$
and suppose $z^3 = 2 + 2i$

then $(r \operatorname{cis} \theta)^3 = 2\sqrt{2} \operatorname{cis} \frac{\pi}{4}$

$\Rightarrow r^3 \operatorname{cis} 3\theta = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ {by De Moivre's theorem} ✓

Hence $r^3 = 2\sqrt{2}$

$\Rightarrow r = \sqrt{2}$

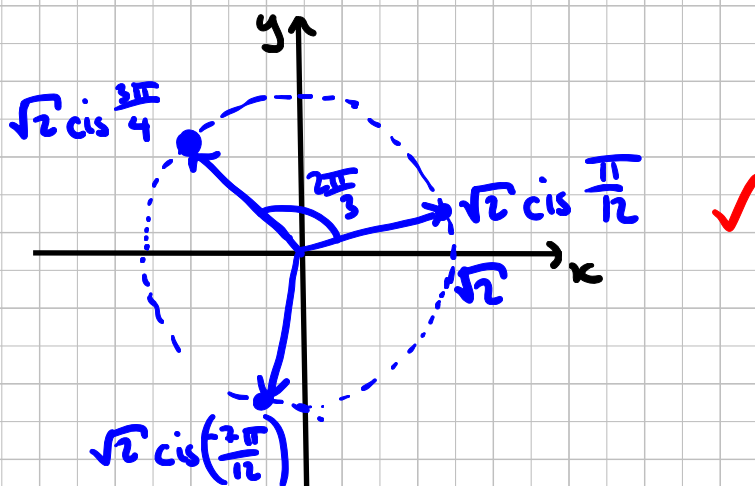
and $3\theta = \frac{\pi}{4} + 2k\pi$

$\theta = \frac{\pi}{12} + \frac{2k\pi}{3}$

Taking $k = 0, \pm 1$ gives

$z = \sqrt{2} \operatorname{cis} \left(\frac{-7\pi}{12} \right), \sqrt{2} \operatorname{cis} \left(\frac{\pi}{12} \right), \sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4} \right)$ ✓

ii)



Let $\omega = \sqrt{2} \operatorname{cis} \frac{\pi}{12}$, then the sum of roots are

$$\sqrt{2} \operatorname{cis} \left(\frac{-7\pi}{12} \right) + \sqrt{2} \operatorname{cis} \left(\frac{\pi}{12} \right) + \sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4} \right)$$

$$= \sqrt{2} \omega \left(\operatorname{cis} \left(-\frac{2\pi}{3} \right) + 1 + \operatorname{cis} \left(\frac{2\pi}{3} \right) \right)$$

$$= \sqrt{2} \omega \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i + 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$= \sqrt{2} \omega \times 0$$

$$= 0 \quad \checkmark$$

15 a) iii)

Adding real parts:

$$\sqrt{2} \cos \frac{\pi}{12} + \sqrt{2} \cos \frac{3\pi}{4} + \sqrt{2} \cos \frac{-7\pi}{12} = 0 \quad \checkmark$$

$$\cos \frac{\pi}{12} - \frac{1}{\sqrt{2}} + \cos \frac{7\pi}{12} = 0$$

$$\cos \frac{\pi}{12} - \cos \frac{5\pi}{12} = \frac{1}{\sqrt{2}}$$

$$\cos \frac{\pi}{12} - \sin \left(\frac{\pi}{2} - \frac{5\pi}{12} \right) = \frac{1}{\sqrt{2}}$$

$$\text{so } \cos \frac{\pi}{12} - \sin \frac{\pi}{12} = \frac{1}{\sqrt{2}}. \quad \checkmark$$

15b) i)

$$P(z) = z^4 + kz^2 + 1$$

Suppose $P(\omega) = 0$;

$$\begin{aligned} \text{then } P(-\omega) &= (-\omega)^4 + k(-\omega)^2 + 1 \\ &= \omega^4 + k\omega^2 + 1 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{and } P(\bar{\omega}) &= (\bar{\omega})^4 + k(\bar{\omega})^2 + 1 \\ &= \overline{\omega^4 + k\omega^2 + 1} \\ &= \overline{\omega^4 + k\omega^2 + 1} \\ &= 0 \end{aligned}$$

(or appeal to
conjugate roots
theorem)

$$\begin{aligned} \text{and } P\left(\frac{1}{\omega}\right) &= \left(\frac{1}{\omega}\right)^4 + k\left(\frac{1}{\omega}\right)^2 + 1 \\ &= \frac{1 + k\omega^2 + \omega^4}{\omega^4} \\ &= 0 \end{aligned}$$

so all are roots

✓ for two
✓✓ for three.

15b) ii)

Since $P(z) = z^4 + kz^2 + 1$,

the product of zeroes is 1.

$$\begin{aligned} \text{but } \omega \times (-\omega) \times \bar{\omega} \times \frac{1}{\bar{\omega}} &= -\omega\bar{\omega} \\ &= -|\omega|^2 \\ &\leq 0 \end{aligned}$$

So these cannot be the four zeroes. ✓

(In fact, you may note that $\omega, -\omega, \bar{\omega}, \frac{1}{\bar{\omega}}, -\bar{\omega}, -\frac{1}{\bar{\omega}}, \frac{1}{\omega}, -\frac{1}{\omega}$ are all zeroes, so some must be doubled up)

Thus at least two of $\omega, -\omega, \bar{\omega}, \frac{1}{\bar{\omega}}$ are equal.

If $|\omega| \neq 1$ then

$\frac{1}{\bar{\omega}}$ cannot be the same as $\omega, \bar{\omega}$, or $-\omega$. ✓

So $\omega = -\omega$ ①

$$\Rightarrow 2\omega = 0$$

$$\Rightarrow \omega = 0 \quad (\text{not possible})$$

$$\omega = \bar{\omega} \quad \text{②}$$

$$\Rightarrow \omega - \bar{\omega} = 0$$

$$\Rightarrow \operatorname{Im}(\omega) = 0$$

or $-\omega = \bar{\omega}$ ③

$$\Rightarrow \omega + \bar{\omega} = 0$$

$$\Rightarrow \operatorname{Re}(\omega) = 0$$



$$15c). i) F = 2000 - mkv^2 \quad \left. \begin{array}{l} m\ddot{x} = 2000 - mkv^2 \\ \ddot{x} = 4 - kv^2 \end{array} \right\} \begin{array}{l} \text{alt. } F = 2000 - kv^2 \\ m\ddot{x} = 2000 - kv^2 \\ \ddot{x} = 4 - \frac{kv^2}{500} \end{array}$$

max speed = 20 m/s so $\ddot{x} = 0$ when $v = 20$

$$\Rightarrow 0 = 4 - 400k$$

$$\Rightarrow k = \frac{1}{100}$$

hence $\ddot{x} = 4 - \frac{v^2}{100}$ ✓

ii) ① v against x :

$$v \frac{dv}{dx} = 4 - \frac{v^2}{100}$$

$$\frac{100v}{400 - v^2} \frac{dv}{dx} = 1$$

$$\int \frac{100v}{400 - v^2} dv = \int dx$$

$$-50 \log|400 - v^2| = x + C$$

when $x = 0$, $v = 10$

$$\text{so } C = -50 \log 300$$

$$\Rightarrow x = 50 \log \left| \frac{300}{400 - v^2} \right| \quad \checkmark$$

when $x = 1000$

$$\frac{300}{400 - v^2} = e^{20}$$

15c) ii) (cont.)

$$400 - v^2 = 300e^{-20}$$

$$v^2 = 400 - 300e^{-20}$$

$$v = \sqrt{400 - 300e^{-20}}$$

② v against t :

$$\ddot{x} = 4 - \frac{v^2}{100}$$

$$\frac{dv}{dt} = \frac{400 - v^2}{100}$$

$$\frac{100}{400 - v^2} \frac{dv}{dt} = 1$$

$$\int \frac{100}{400 - v^2} dv = \int dt$$

$$\frac{100}{40} \int \frac{1}{20+v} + \frac{1}{20-v} dv = t + C$$

$$\frac{5}{2} \log \left| \frac{20+v}{20-v} \right| = t + C$$

$$\text{when } t = 0, v = 10$$

$$\text{so } C = \frac{5}{2} \log |3|$$

$$\text{Hence } t = \frac{5}{2} \log \left| \frac{20+v}{20-3v} \right|$$

③ combining:

$$\text{Letting } v = \sqrt{400 - 300e^{-20}}$$

$$\text{gives } t = 51.438 \text{ seconds}$$

$$\approx 51 \text{ seconds}$$

16a)

RTP:

$$n!! = \frac{n!}{2^{\frac{n-1}{2}} \times \left(\frac{n-1}{2}\right)!} \quad \text{for } n \text{ odd}$$

A

when $n=1$

$$\begin{aligned} \text{LHS} &= 1!! \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{1!}{2^0 \times 0!} \\ &= 1 \end{aligned}$$

so $\text{LHS} = \text{RHS}$ Thus the result holds for $n=1$ ✓B Assume the result holds for $n=k$, for k odd.Now for $n=k+2$,

$$(k+2)!! = (k+2) \times k!!$$

$$= (k+2) \times \frac{k!}{2^{\frac{k-1}{2}} \times \left(\frac{k-1}{2}\right)!} \quad \left\{ \begin{array}{l} \text{By our} \\ \text{assumption} \end{array} \right\}$$

$$= \frac{(k+2) \times (k+1) \times k!}{(k+1) \times 2^{\frac{k-1}{2}} \times \left(\frac{k-1}{2}\right)!}$$

$$= \frac{(k+2)!}{2 \times \left(\frac{k+1}{2}\right) \times 2^{\frac{k-1}{2}} \times \left(\frac{k-1}{2}\right)!}$$

$$= \frac{(k+2)!}{2^{\frac{k+1}{2}} \times \left(\frac{k+1}{2}\right)!} \quad \checkmark$$

Thus the result holds for $n=k+2$ whenever it holds for $n=k$ C So by parts A & B, the result holds for all $n \geq 1$ odd.

16b) i)

$$\text{Let } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx$$

$$\text{let } x = \frac{\pi}{2} - u$$

$$dx = -du$$

$$\text{when } x=0, u = \frac{\pi}{2}$$

$$x = \frac{\pi}{2}, u = 0$$

$$\text{so } I_{m,n} = \int_{\frac{\pi}{2}}^0 \sin^m \left(\frac{\pi}{2} - u \right) \cos^n \left(\frac{\pi}{2} - u \right) (-du)$$

$$= \int_0^{\frac{\pi}{2}} \cos^m(u) \sin^n(u) \, du \quad \checkmark$$

$$= \int_0^{\frac{\pi}{2}} \sin^n(u) \cos^m(u) \, du$$

$$= I_{n,m}$$

ii)

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx$$

$$= \int_0^{\frac{\pi}{2}} (\cos^{n-1} x) (\sin^m x \cos x \, dx)$$

$$= \left[\frac{1}{m+1} \cos^{n-1} x \sin^{m+1} x \right]_0^{\frac{\pi}{2}}$$

$$- \int_0^{\frac{\pi}{2}} \frac{n-1}{m+1} \sin^{m+1} x (\cos^{n-2} x \cdot -\sin x) \, dx$$

$$\text{Let } u = \cos^{n-1} x$$

$$du = (n-1) \cos^{n-2} x$$

$$\cdot (-\sin x) \, dx$$

$$dv = \sin^m x \cos x \, dx$$

$$v = \frac{1}{m+1} \sin^{m+1} x \quad \checkmark$$

$$= [0 - 0] + \frac{n-1}{m+1} \int_0^{\frac{\pi}{2}} \sin^{m+2} x \cos^{n-2} x \, dx$$

$$= \frac{n-1}{m+1} \int_0^{\frac{\pi}{2}} \sin^m x (1 - \cos^2 x) \cos^{n-2} x \, dx \quad \checkmark$$

1bb) ii) (cont.)

$$I_{m,n} = \frac{n-1}{m+1} [I_{m,n-2} - I_{m,n}]$$

$$(m+1)I_{m,n} = (n-1)I_{m,n-2} - (n-1)I_{m,n}$$

$$(m+n)I_{m,n} = (n-1)I_{m,n-2}$$

$$I_{m,n} = \frac{n-1}{m+n} I_{m,n-2} \quad \text{as required.} \quad \checkmark$$

$$\text{iii) } I_{2k,2l} = \frac{(2l-1)}{(2k+2l)} I_{2k,2l-2} \quad (\text{by part ii})$$

$$= \frac{(2l-1)}{2(k+l)} \times \frac{(2l-3)}{2(k+l-1)} I_{2k,2l-4}$$

$$= \frac{(2l-1)}{2(k+l)} \times \frac{(2l-3)}{2(k+l-1)} \times \dots \times \frac{1}{2(k+1)} I_{2k,0} \quad \checkmark$$

$$I_{2k,0} = I_{0,2k} \quad (\text{by part i})$$

$$= \frac{(2k-1)}{2k} \times \frac{(2k-3)}{2(k-1)} \times \dots \times \frac{1}{2} I_{0,0}$$

$$\text{so } I_{2k,2l} = \frac{(2l-1)!! \times (2k-1)!!}{2^{k+l} (k+l)!} I_{0,0} \quad \checkmark$$

$$= \frac{(2l-1)!}{2^{l-1} \times (l-1)!} \times \frac{(2k-1)!}{2^{k-1} \times (k-1)!} \times \frac{1}{2^{k+l} \times (k+l)!} \times I_{0,0}$$

{using 1b a}

16b) iii) (cont.)

$$I_{2k,2\ell} = \frac{(2\ell)!}{2^\ell \times \ell!} \times \frac{(2k)!}{2^k \times k!} \times \frac{1}{2^{k+\ell} \times (k+\ell)!} \times I_{0,0}$$

$$= \frac{(2\ell)! (2k)!}{(\ell)! (k)! (k+\ell)!} \times \frac{I_{0,0}}{2^{2k+2\ell}}$$

$$I_{0,0} = \int_0^{\frac{\pi}{2}} dx \\ = \frac{\pi}{2}$$

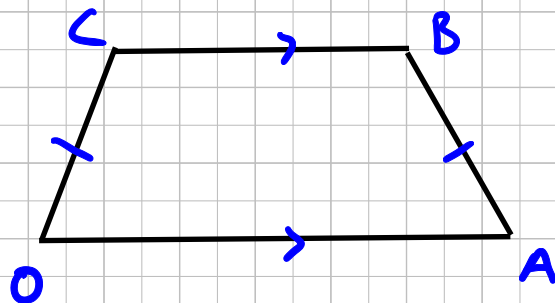
$$\text{So } I_{2k,2\ell} = \frac{(2\ell)! (2k)!}{\ell! k! (k+\ell)!} \times \frac{\pi}{2^{2k+2\ell+1}} \text{ as req'd.}$$



16c) Let $\underline{a} = \overrightarrow{OA}$, $\underline{c} = \overrightarrow{OC}$

$\overrightarrow{CB} \parallel \overrightarrow{OA}$ so

$\overrightarrow{CB} = k \underline{a}$ for some $k \neq 1$.



$$\begin{aligned}\text{Then } \overrightarrow{AB} &= -\underline{a} + \underline{c} + k \underline{a} \\ &= (k-1) \underline{a} + \underline{c} \quad \checkmark\end{aligned}$$

$$|\overrightarrow{OC}| = |\overrightarrow{AB}| \quad \text{so} \quad |\overrightarrow{OC}|^2 = |\overrightarrow{AB}|^2$$

$$\begin{aligned}|\underline{c}|^2 &= ((k-1) \underline{a} + \underline{c}) \cdot ((k-1) \underline{a} + \underline{c}) \\ &= (k-1)^2 |\underline{a}|^2 + 2(k-1) \underline{a} \cdot \underline{c} + |\underline{c}|^2\end{aligned}$$

$$\Rightarrow 0 = (k-1)^2 |\underline{a}|^2 + 2(k-1) \underline{a} \cdot \underline{c}$$

$$(k-1) \neq 0$$

$$\begin{aligned}\text{so } (k-1) |\underline{a}|^2 + 2 \underline{a} \cdot \underline{c} &= 0 \\ \Rightarrow 2 \underline{a} \cdot \underline{c} &= (1-k) |\underline{a}|^2 \quad \checkmark\end{aligned}$$

$$\text{Now } \overrightarrow{OB} = \underline{c} + k \underline{a}$$

$$\text{and } \overrightarrow{AC} = \underline{c} - \underline{a}$$

$$\begin{aligned}|\overrightarrow{OB}|^2 &= |\underline{c}|^2 + 2k \underline{a} \cdot \underline{c} + k^2 |\underline{a}|^2 \\ &= |\underline{c}|^2 + k(1-k) |\underline{a}|^2 + k^2 |\underline{a}|^2 \\ &= |\underline{c}|^2 + k |\underline{a}|^2\end{aligned}$$

$$\begin{aligned}|\overrightarrow{AC}|^2 &= |\underline{c}|^2 - 2 \underline{a} \cdot \underline{c} + |\underline{a}|^2 \\ &= |\underline{c}|^2 - (1-k) |\underline{a}|^2 + |\underline{a}|^2 \\ &= |\underline{c}|^2 + k |\underline{a}|^2 \quad \checkmark\end{aligned}$$

$$= |\overrightarrow{OB}|^2$$

so $|\overrightarrow{AC}| = |\overrightarrow{OB}|$ as required.